Fundamentals of Optimization: Formulations, Solution Spaces, and Combinatorial Models

An overview defining the core components, mathematical formulations, algorithmic classifications, and applied models of mathematical optimization.

THE MATHEMATICAL BLUEPRINT: OPTIMIZATION ROADMAP INDEX

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1. Foundations of Optimization Models

Motivation: To formulate mathematical solutions, we must first define the universal components that constitute any optimization problem.

2. Feasibility, Optimality, and Solution Spaces

3. Taxonomy: Continuous vs. Discrete Optimization

4. Combinatorial Optimization and Linear Programming Formulations

Optimization: The selection of a best element (with regard to some criterion) from some set of available alternatives.

Classical Example: $\max w_1x_1 + w_2x_2$ s.t. $x_1 + x_2 \leq B$ $(x_1, x_2 \geq 0)$

Decision Variables
$(\mathbf{x} \in \mathbb{R}^n)$

The variables that the decision maker can control or choose.

Objective Function
$(f(\mathbf{x}))$

A mathematical expression that maps choices to a scalar value measuring performance (e.g., $w_1x_1 + w_2x_2$ as profit) to be minimized or maximized.

Constraints
$(g_i(\mathbf{x}) \leq 0, h_j(\mathbf{x}) = 0)$

Restrictions or conditions that must be satisfied by the decision variables.

Continuous Optimization

Variables take continuous values from a range (e.g., $x \in \mathbb{R}^n$). Feasible region is often a continuous set.

$$\min_{x \in S} f(x)$$

$$\text{s.t.} \quad h_i(x) = 0, i = 1, \ldots, m,$$
$$g_j(x) \leq 0, j = 1, \ldots, p$$

Note: $x \in \mathbb{R}^n$ is the decision vector (Teal), f the objective function (Gold), h_i, g_j constraint functions (Crimson), and S a given set (Teal). A linear program is a special case where f and all constraints are linear.

Discrete Optimization

Variables are restricted to specific, distinct values (e.g., integers, $x \in \mathbb{Z}^n$ or binary $x \in \{0, 1\}^n$). Feasible region is a set of distinct points.

Minimize: Find feasible x^* where $f(x^*) \leq f(x)\ \forall x$ feasible.

Conversion Rule: Any maximization can be converted to minimization by minimizing $-f(x)$, and vice versa.

Maximize: Find feasible x^* where $f(x^*) \geq f(x)\ \forall x$ feasible.

1. Problem Formulation

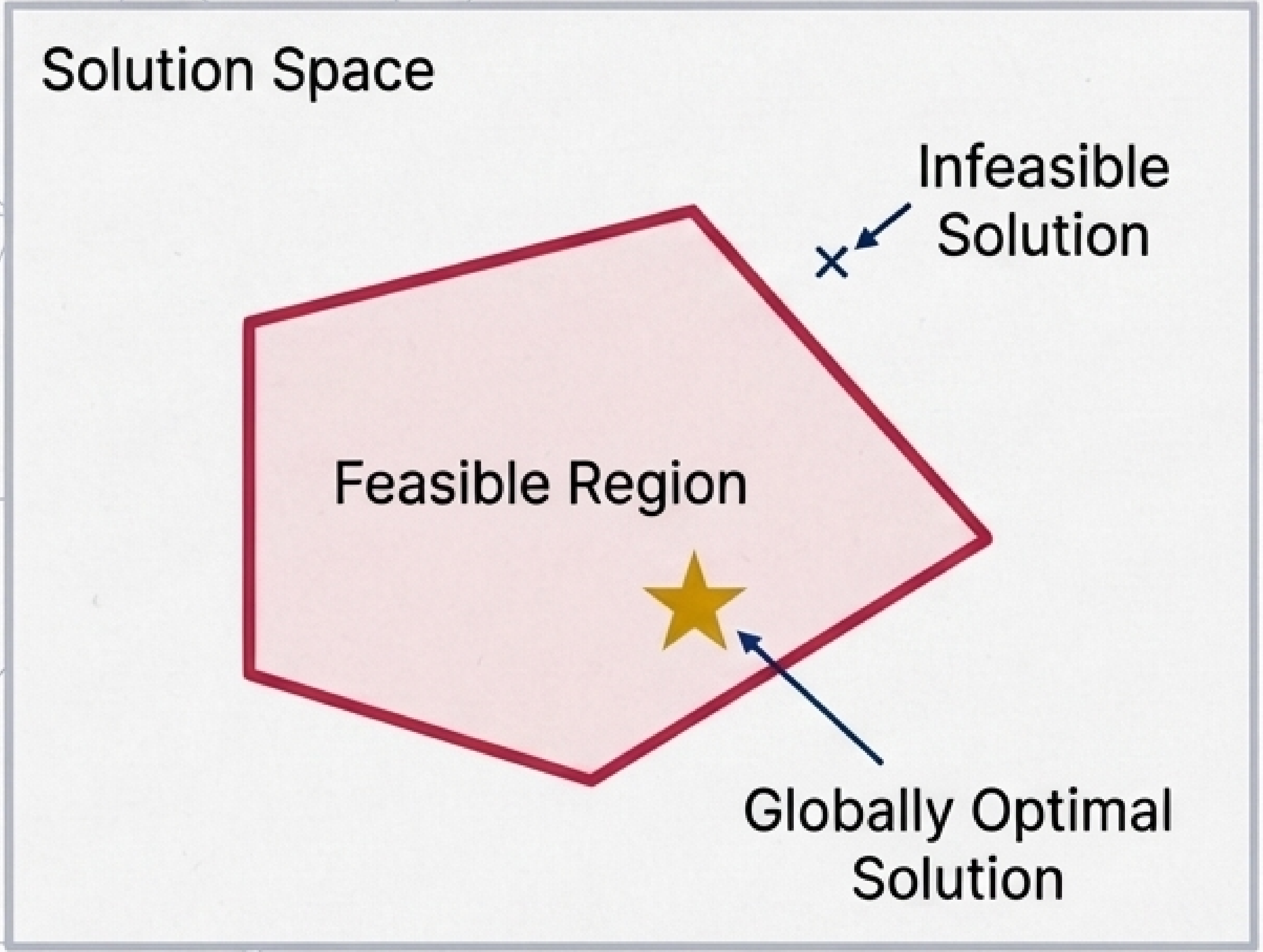
2. Feasibility, Optimality, and Solution Spaces

3. Algorithms & Methods

4. Practical Considerations

Motivation:

Having established the mathematical formulation, we must now define how constraints bound the variables to create distinct spaces of potential and valid solution space.



Solution Space (Decision Space): The full set of all possible values variables can take before enforcing constraints (e.g., subset of $\mathbb{R}^n$ or $\{0,1\}^n$).

Feasible Region (Feasible Set): The collection of all decision vectors satisfying **all constraints**.

Feasible vs. Infeasible Solutions: Feasible solutions satisfy all constraints. Infeasible solutions violate at least one constraint.

Globally Optimal Solution (x^*): A feasible solution whose objective value is the absolute best among all feasible solutions over the feasible set.

As Example

Find the solution space and feasible solution space for the given binary optimization problem:

$$\begin{aligned} \max \quad & 5x_1 + 7x_2 + 2x_3 \\ \text{s.t.} \quad & 2x_1 + 4x_2 + x_3 \leq 4, \\ & x_1, x_2, x_3 \in \{0, 1\}. \end{aligned}$$

Component Analysis

Objective: Maximize the objective function.

Decision Variables: Binary variables x_1, x_2, x_3.

Constraint: A single resource limitation inequality.

Solution Space (X): $X = \{0,1\}^3 = \{(x_1, x_2, x_3) \mid x_i \in \{0,1\}, i = 1,2,3\}$. Contains 2^3 possible (feasible + infeasible) = 8 possible discrete points.

Constraint: $2x_1 + 4x_2 + x_3 \leq 4$ for $x \in \{0,1\}^3$

Incumbent Solution

Initial Incumbent: None (or worst possible).
Update incumbent if current feasible solution is better during [Add to X_F].

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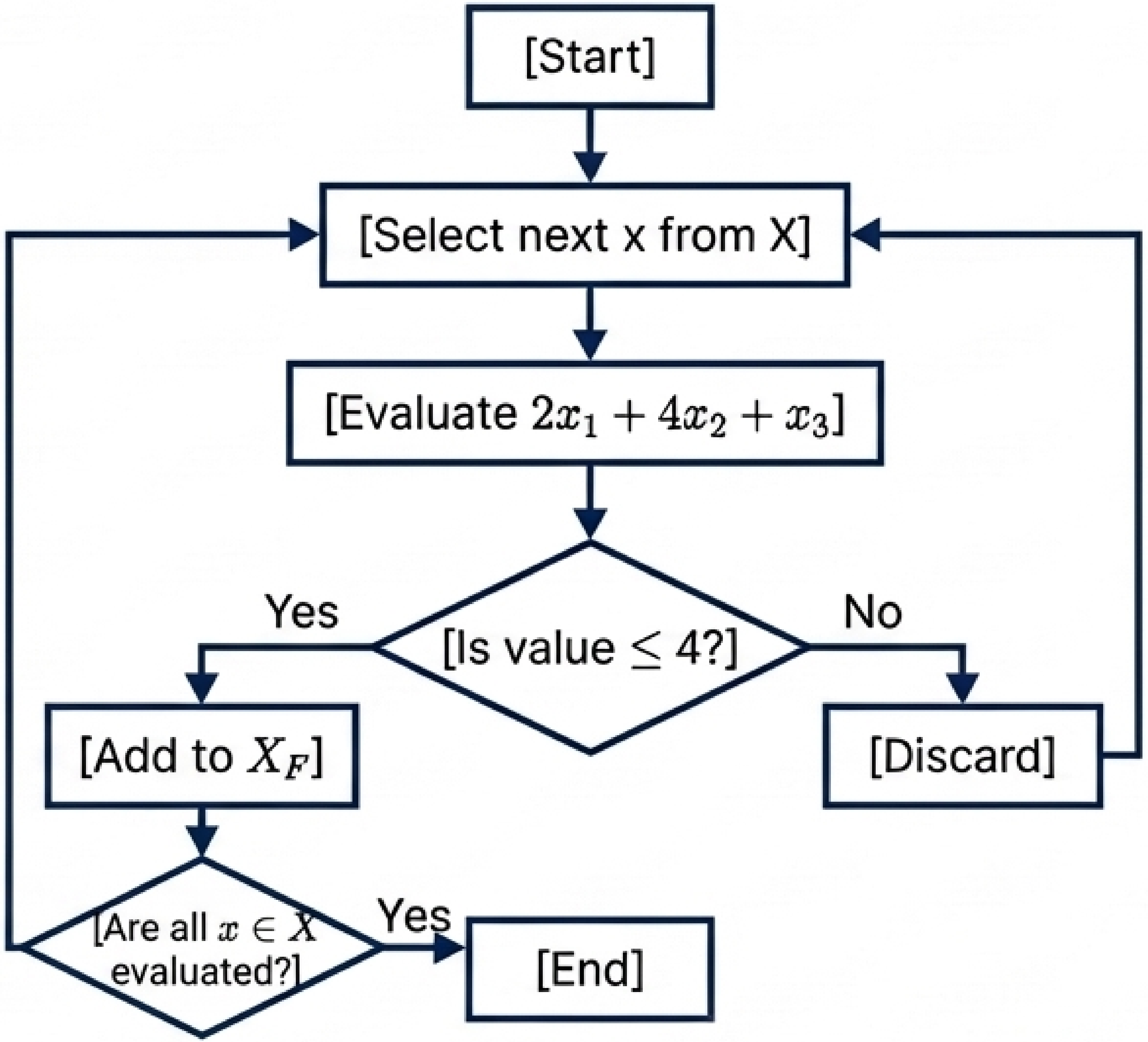
X_F = empty set
For each point x in { (0,0,0) to (1,1,1) }:
    value = 2*x_1 + 4*x_2 + x_3
    If value <= 4:
        add x to X_F
Return X_F
```



Result (Feasible Solution Space X_F): Output subset is exactly $\{(0,0,0), (0,0,1), (0,1,0), (1,0,0), (1,0,1)\}$. (Points $(0,1,1)$, $(1,1,0)$, $(1,1,1)$ are discarded as infeasible).

1. Problem Formulation

2. Solution Space Analysis

3. Taxonomy: Continuous vs. Discrete Optimization

4. Algorithm Selection

Motivation: Because **solution spaces** vary drastically—from infinite real numbers to finite binary sets—**optimization problems** require distinct mathematical classifications and algorithmic approaches.

Continuous / Calculus-Based Optimization	• **Characteristics:** Variables take real values ($x \in \mathbb{R}^n$). Differentiable objective and constraint functions. • **Linear Form:** Affine objective ($f(x) = c^T x$); no products/powers beyond first degree. • **Nonlinear Form:** Contains products ($x_i x_j$), powers (x_i^2), or exponentials. • **Algorithmic Methods:** Gradient, Hessian, KKT conditions, Newton, and quasi-Newton methods.
Discrete / Combinatorial Optimization	• **Characteristics:** Variables restricted to discrete sets (integers, binaries $\{0, 1\}^n$). Feasible set is finite/countable; derivatives are not defined. Deals with graphs, sets, or integer vectors. • **Linear Form:** Objective and constraints are linear functions of discrete variables. • **Algorithmic Methods:** Branch-and-bound, cutting planes, dynamic programming, network flows.

1. Problem Formulation

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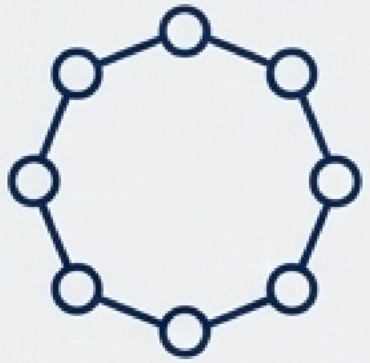
4. Combinatorial Optimization and Linear Programming Formulations

Motivation: Theoretical combinatorial structures form the backbone of algorithms used to solve complex logistical and operational challenges in the real world.

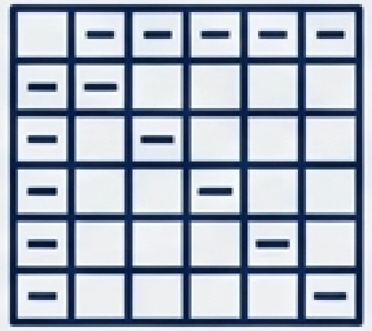


Traveling Salesman Problem (TSP)

Find the shortest tour visiting each city exactly once and returning to the start.

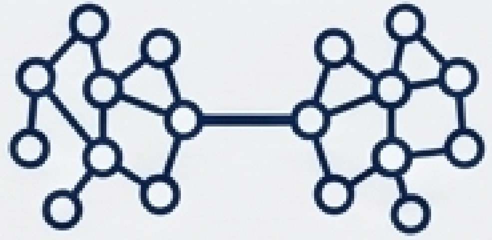
Vehicle Routing and Logistics

Design routes for delivery trucks to minimize total distance or cost under capacity constraints.



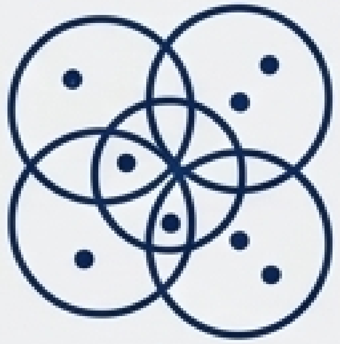
Crew and Shift Scheduling

Assign staff to shifts or flights while respecting legal/contractual rules and minimizing cost.



Network Design

Choose which links to build in a communication or power network to satisfy demand at minimum cost.



Set Covering / Facility Location

Select facilities or sets to cover demand points with minimal total cost.

Combinatorial, Linear, and Non-Linear Optimization

Combinatorial Optimization

The process of finding an optimal object from a finite set of objects. The domain of the objective function is discrete.

Linear Programming

A method to achieve the best outcome (such as maximum profit or lowest cost) in a mathematical model whose requirements are represented by linear relationships.

Non-Linear Programming

The process of solving an optimization problem where some of the constraints or the objective function are non-linear.

Synthesis and Conclusion

Optimization fundamentally relies on defining decision variables, establishing a objective objective function, and delineating a bounded feasible region through strict mathematical constraints.

The transition from theoretical solution spaces to feasible spaces can be algorithmically evaluated, filtering mathematical possibility into operational reality.

Problem topology dictates the methodology: continuous formulations utilize calculus-based methods, whereas discrete, combinatorial structures require specialized integer programming to solve complex real-world logistical networks.